

VIT BHOPAL UNIVERSITY

School of Computer Science and Engineering

PROJECT REPORT

on

Contact Book – A Command-Line Application

(Developed in Python)

Project Title	Contact Book (CLI)
Language / Platform	Python 3 (standard library only, no external dependencies)
Project Type	Command-line application
Submitted By	Rushikesh Ganesh Pabale
Registration Number	26BCE10724
Program / Branch	B.Tech
Course / Module	CSE1021
Faculty Guide	Pradeep Kumar Mishra
Institution	VIT Bhopal University
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1. Introduction

The Contact Book is a command-line application built in Python that allows a user to store, retrieve, update, and delete personal contact information during a single program session. It was built to demonstrate core programming concepts such as functions, dictionaries, control flow, input validation, and a menu-driven program structure, without relying on any external libraries or frameworks.

2. Objective

The objective of this project is to design and implement a simple, fully functional contact management tool that runs entirely from the terminal. The program is intended to:

- Allow users to add new contacts with a name, phone number, and optional email.
- Allow users to view all saved contacts in alphabetical order.
- Allow users to search for a specific contact by name.
- Allow users to update an existing contact's details.
- Allow users to delete a contact, with confirmation before removal.
- Present all of this through a simple, numbered, menu-driven command-line interface.

3. Tools and Technologies Used

Programming Language	Python 3 (developed and tested on Python 3.12)
Libraries Used	None — only Python's built-in features (dict, input, print, str methods)
Development Environment	Any text editor / IDE + a terminal (e.g. VS Code, PyCharm, or plain terminal)
Execution Environment	Command line / terminal (Windows, macOS, or Linux)

4. System Design / Methodology

The application follows a simple procedural design centered on a single in-memory data structure and a set of functions, each responsible for one menu operation. There is no database or file storage in this version — all data lives in memory for the duration of the program's execution.

4.1 Data Structure

Contacts are stored in a single dictionary named `contacts`, where each key is a contact's name and each value is another dictionary holding that contact's phone and email:

```
contacts = { "Jane Doe": {"phone": "9876543210", "email": "jane@example.com"} }
```

4.2 Program Flow

- The program starts by calling `programm()`, which runs an infinite loop.
- On each loop iteration, `menu()` prints the six available options.
- The user's choice is read and matched against options 1–6 using `if/elif` statements.
- Each option calls its corresponding function: `add()`, `view()`, `search()`, `update()`, or `delete()`.
- Choosing option 6 prints an exit message and breaks out of the loop, ending the program.
- Any input outside 1–6 triggers an 'Invalid choice' message and redisplay the menu.

5. Features and Functionality

5.1 Add Contact

Prompts the user for a name, phone number, and an optional email. If the name already exists in the dictionary, the program refuses to add a duplicate and suggests using the update option instead.

5.2 View All Contacts

Iterates over the contacts dictionary sorted alphabetically by name and prints each contact's name, phone, and email (showing 'N/A' if no email was provided). If no contacts exist yet, it informs the user instead of printing an empty list.

5.3 Search Contact

Looks up a contact by exact, case-sensitive name match and displays its details, or reports that no matching contact was found.

5.4 Update Contact

Looks up a contact by name and lets the user enter new phone and/or email values. Leaving a field blank keeps the existing value unchanged, so partial updates are supported.

5.5 Delete Contact

Looks up a contact by name and asks for confirmation ('y' to proceed) before permanently removing it from the dictionary, to guard against accidental deletion.

6. Code Structure

The program is organized into six functions plus a driver loop, all in a single file (`contact_book.py`):

- `add()` — creates a new contact entry
- `search()` — finds and displays one contact
- `view()` — displays all contacts, sorted
- `delete()` — removes a contact after confirmation
- `update()` — edits an existing contact's phone/email
- `menu()` — prints the numbered menu
- `programm()` — the main loop that ties everything together

7. How to Run the Project

This project is fully executable from the command line and requires no GUI setup:

- Ensure Python 3.6+ is installed (`python3 --version`).
- Place `contact_book.py` in a working folder.
- Open a terminal in that folder.
- Run: `python3 contact_book.py` (or `python contact_book.py` on some Windows setups).
- Interact with the numbered menu (1–6) using the keyboard.

Full setup and usage instructions are provided in the accompanying `README.md` file.

8. Sample Output

Below is a representative example of the program's menu as seen in the terminal:

```
===== CONTACT BOOK =====  
1. Add Contact  
2. View All Contacts  
3. Search Contact  
4. Update Contact  
5. Delete Contact  
6. Exit  
Enter your choice (1-6):
```

9. Testing

The program was manually tested by running it in a terminal and exercising each menu option in turn: adding contacts (including attempting a duplicate name), viewing the list before and after changes, searching for both existing and non-existing names, updating a contact with partial and full field changes, deleting a contact with both confirmation and cancellation, and entering invalid menu choices to confirm error handling.

10. Limitations and Future Scope

- No persistence: all contacts are lost when the program exits. A future version could save/load data to a JSON or CSV file, or a database.
- Name matching is exact and case-sensitive; future versions could support partial or case-insensitive search.
- No validation of phone number or email format is currently performed.
- A graphical or web-based interface could be added on top of the same core logic.

11. Conclusion

The Contact Book project successfully demonstrates a complete, menu-driven command-line application built using core Python concepts. It fulfills its stated objective of allowing users to add, view, search, update, and delete contacts through a straightforward terminal interface, and it is fully executable via the command line without any GUI dependency.